

Chapter 2

Your computer (Hardware)

You need to know what's inside your computer, and truly understanding how it works is key to understanding everything else. For me, this knowledge is essential and should be second nature.

2.1 The Essentials

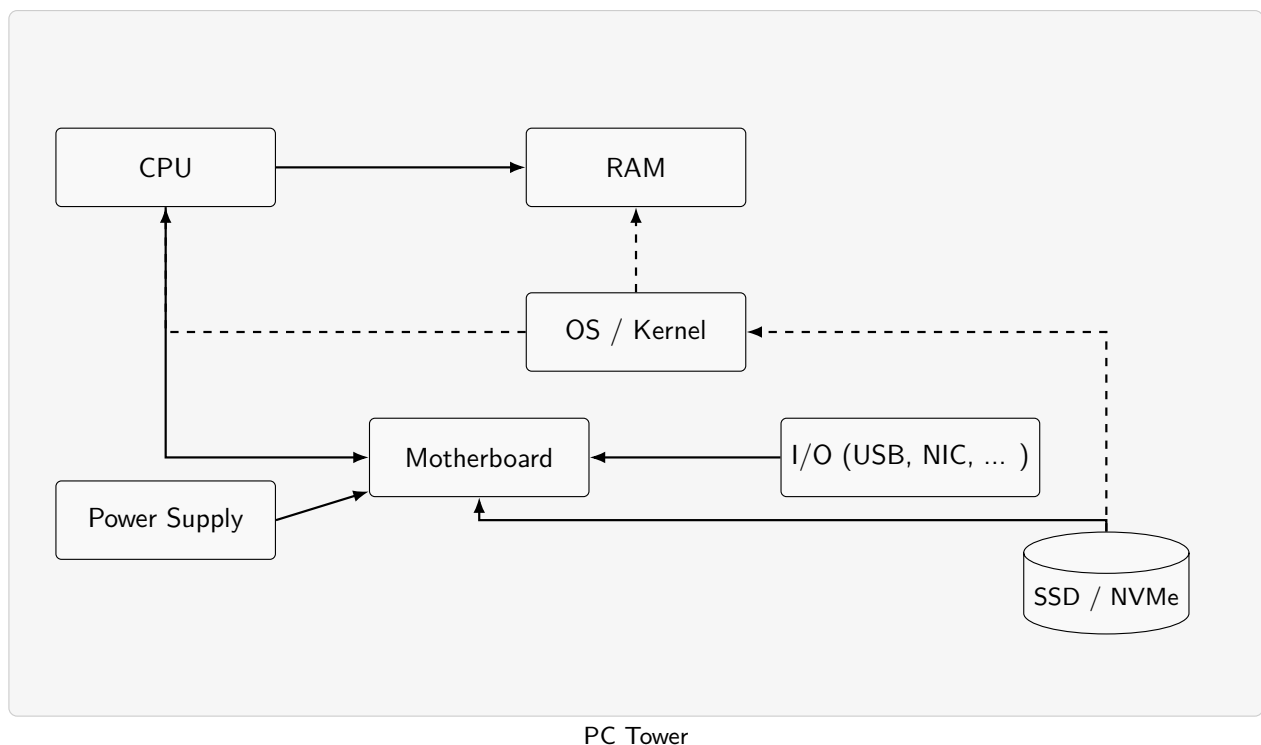


Figure 2.1: Computer essentials (espacé, flèches sans croisement)

2.1.1 Processor (CPU - Central Processing Unit)

The **processor** is often described as the "brain" of the computer. It executes program instructions and performs the necessary calculations. When you run a program (e.g., from a terminal), it is the processor that interprets and executes machine instructions.

Operation:

- **Interprets and executes instructions:** The processor takes instructions in the form of machine code from the OS (binary) and executes them sequentially.
- **Performs calculations:** It carries out all the necessary arithmetic and logical operations.

- **Process management:** The processor manages multiple processes simultaneously using features such as multitasking or multi-threading (parallel execution).

2.1.2 Random Access Memory (RAM)

Role:

RAM is short-term memory used to temporarily store data and programs during execution. The more RAM you have, the more tasks the computer can handle simultaneously without slowing down.

Operation:

- **Temporary data storage:** When you launch a program, it is loaded from the hard drive into RAM to allow for quick access.
- **Fast execution:** The processor accesses data in RAM much faster than if it had to read it directly from the hard drive.
- **Multitasking:** RAM stores multiple applications and processes running simultaneously, which is essential for multitasking.

2.1.3 Disk Space (HDD/SSD - Hard Disk Drive/Solid State Drive)

Role:

Disk space is used for permanent data and program storage. Traditional hard drives (HDD) are slower than **SSDs**, which are much faster flash memories.

Operation:

- **Permanent storage:** Files, operating systems, applications, and user data are stored permanently on the disk until they are deleted.
- **Program access and loading:** When a program is launched, its files are first retrieved from the hard drive/SSD and then loaded into RAM to be used by the processor.
- **Read/write speed:** SSDs allow much faster data access than HDDs, speeding up program startup and file loading.

Aside: Data Storage in RAM and SSD

Data Storage Principles:

Data in both RAM and SSDs are stored using a structured system of memory "cells" or "blocks" that hold binary information (0s and 1s). These cells are addressed in sequences, allowing data to be accessed or written in an organized manner. Despite their similarities, RAM and SSDs have distinct architectures suited to different types of operations.

RAM Data Storage:

- **Sequential Storage and Access:** RAM (Random Access Memory) is divided into a series of cells, each with a unique address, enabling quick, random access to any stored data. When data is stored in RAM, the system fills these cells sequentially, and each piece of information has an address that the CPU can use to retrieve or modify it rapidly.
- **Optimization and Management:** Data in RAM is volatile (it is erased when power is lost), which makes it ideal for temporary storage during active processes. The operating system and the memory management unit (MMU) ensure efficient usage by dynamically allocating and deallocating memory as programs require, avoiding "fragmentation" (wasted memory space).
- **UTF-8 Encoding and Data Representation:** Textual data is often stored using UTF-8, a standard encoding system that converts characters into binary. UTF-8 allows flexible, variable-length encoding, efficiently representing text and supporting a wide range of symbols and languages while minimizing space usage.

SSD Data Storage:

- **Block Structure and Sequential Writing:** SSDs store data in blocks composed of smaller units called pages. Unlike RAM, SSDs cannot overwrite data in place, so data is stored sequentially in available blocks, and blocks must be erased before they are rewritten. This design allows SSDs to maintain high read speeds but complicates write operations.
- **Wear-Leveling and Optimization:** Since each block has a limited number of writes, SSDs use wear-leveling algorithms to spread data evenly across the storage cells, extending the lifespan of the drive. This ensures optimal utilization of the storage space and helps prevent early "wear-out" of frequently written areas.
- **Addressing and Encoding:** Like RAM, SSDs use addressing to manage data access, though they handle it at the block level rather than at the byte level. UTF-8 encoding is also used for textual data, allowing efficient storage of character-based information with variable-length encoding that saves space and supports diverse character sets.

In summary, while both RAM and SSDs utilize sequential filling and addressing to manage data, their specific storage mechanisms and optimizations vary to align with their roles—RAM prioritizes speed and volatility, while SSDs focus on durability and permanent storage management.

2.1.4 Motherboard**Role:**

The **motherboard** is the main printed circuit board of the computer that connects all the components together. It facilitates communication between the processor, RAM, hard drive, and all other computer components.

Operation:

- **Connectivity:** It physically connects all components (CPU, RAM, GPU, storage) and allows their communication via data buses.
- **Peripheral management:** The motherboard also allows connection of external peripherals (keyboard, mouse, etc.) and expansion cards (graphics cards, sound cards, etc.).
- **Power distribution:** It distributes electricity to the various components from the power supply (PSU).

2.1.5 System Bus**Role:**

The **system bus** is a set of communication lines that allow different components of the computer to communicate with each other.

Operation:

- **Data bus:** Transports data between components (e.g., between the processor and RAM).
- **Address bus:** Transports memory addresses to identify where data should be read or written.
- **Control bus:** Transports control signals, which coordinate operations between components.

2.1.6 Operating System (OS)**Role:**

The **operating system (OS)** is software that manages the computer's hardware and software resources. It allows the user and applications to interact with the hardware in a simplified way.

Operation:

- **Process management:** The OS manages running processes, allocates resources (like RAM and CPU), and ensures multitasking.
- **User interface:** It provides a user interface (graphical or command-line) to interact with the system.
- **Device management:** It manages communication between the computer and external devices (keyboards, printers, etc.).

2.2 Some Others

Role: The **graphics card** or **GPU** is primarily responsible for rendering visuals and managing graphics. For certain intensive applications (like video games or design software), the GPU can also perform complex calculations in parallel with the processor.

Operation:

- **Graphics rendering:** The GPU processes graphical information (images, videos, animations) and displays it on the screen.
- **Parallel computation:** The GPU can also be used for massive parallel computations (such as image processing or AI applications).

2.2.1 Virtual Memory

Virtual Memory:

Virtual memory is a mechanism used by the operating system to compensate for a lack of RAM by using part of the disk space to temporarily store data.

Operation:

- **Swapping:** When the RAM is full, the operating system moves some data from RAM to a space on the disk called the **swap file** or **pagefile**.
- **Less efficient than RAM:** Accessing data in virtual memory is slower than accessing data in RAM, as the hard drive/SSD is much slower than the RAM.

2.2.2 Power Supply Unit (PSU)

Role:

The **power supply unit (PSU)** provides the necessary energy for all computer components. It converts electricity from the mains into usable current for internal components.

Operation:

- **Power distribution:** The PSU distributes power in the form of different voltages (typically 3.3V, 5V, and 12V) to components such as the processor, graphics card, hard drives, and motherboard.
- **Power stability:** A low-quality power supply can cause malfunctions, crashes, or even hardware damage.

2.2.3 Memory Management Unit (MMU)

Role:

The **memory management unit (MMU)** is responsible for handling all memory access requests made by the CPU. It manages memory addresses, translating virtual addresses to physical addresses and ensuring efficient, secure access to the memory.

Operation: